RAGs for Open Domain Complex QA

Samuel Karskens
s.m.f.karskens@student.tudelft.nl

Vithursika Vinasiththamby
vvinasiththamb@student.tudelft.nl

Sankalp Sagar
sankalpsagar@tudelft.nl

Maria Elvira Antonogiannaki
M.E.Antonogiannaki@student.tudelft.nl

Abstract

Question-Answering problems struggle with providing reliable and verified answers when answers are generated by Large Language Models (LLM) since it is prone to hallucinations. Retrieval-Augmented Generation can provide a solution by retrieving the relevant external documents. This paper explores four different types of contexts, namely oracle, related, random negatives and hard negatives, on Open Domain Complex QA. We use the 2WikiMultiHopQA dataset and evaluate our performance using the Exact Match (EM) metric for different retrieval strategies such as Contriever and ADORE. We find that the inclusion of random negatives improves the performance of the LLM, while relying on hard negatives can confuse the LLM. Furthermore, Contriever performs better than ADORE in most settings.

1 Introduction

Obtaining valid answers to questions is a critical point in many fields, for example in education. Recently, Large Language Models (LLMs) have been proposed as a solution to this problem as they can generate text that can pass to be written by a human. However, direct prompting of the LLM will not always result in a proper and valid answer as the question might require the LLM to have knowledge about specific domains. Additionally, direct prompting can also result in a higher probability of hallucinations — responses that sound possible but are factually incorrect. (Ji et al., 2022)

A possible solution to this problem is Retrieval-Augmented Generation (RAG), which combines LLMs with retrieval machines. RAG enhances the quality of the answer by incorporating additional external information in the prompt. (Gao et al., 2024) Furthermore, Few-Shot prompting can be utilised where multiple input-output examples are given to the model, such that it can learn the expected format for the answers.

In-Context Learning (ICL) is a technique that allows the LLM to adapt to unseen domains by learning from the context of the input alone. (Xie et al., 2021) The benefit of this approach is that it does not require any further fine-tuning to adapt the LLM to the new domain.

Several researchers have worked on identifying the type of context that can enhance the performance of a RAG system in providing the correct responses. For example, (Cuconasu et al., 2024) argued that the retrieval component of RAGs should be further investigated, in order to better understand which characteristics could optimize the prompts used as input in the retriever. To address that, they conducted a study examining the types of passages that should be retrieved. They discovered that injecting random documents in the prompt enhances the ability of LLMs to generate correct outputs.

In this research, we conduct a systematic study that investigates the impact of diverse types of contexts in the quality of answer generation of RAG systems for complex Question Answering (QA). We evaluate the performance of the different contexts on the compositional questions from 2WikiMultiHopQA (Ho et al., 2020) by running several experiments, addressing the following research questions:

- How does negative contexts impact downstream answer generation performance?
- Are negative context more important for answer generation than related contexts?
- Does providing only gold contexts deteriorate the performance compared to mixing with other negative or related contexts?

The remainder of this paper is structured as follows. In Section 2, we discuss related work to Retrieval-Augmented Generation (RAG) and Multihop Question Answering. Section 3 describes the methodology and the experimental setup and

in Section 4, we present and analyze our findings to the research questions. A qualitative analysis and discussion can be found in Section 5 and 6, respectively. Finally, the conclusion of our research can be found in Section 7. The code used for our experiments is available in our Github repository (Karskens et al., 2024).

2 Related Work

2.1 Question Answering

To tackle complex QA problems Large Language Models (LLMs) are often used. (Xie et al., 2021) discusses the phenomenon of In-Context Learning (ICL) in LLMs. ICL occurs when the LLM is presented with a few examples of input and output such that the model does not require any explicit pre-training, and can perform the task at hand well. The authors of this paper understand this as a form of Bayesian inference. The LLM updates its 'beliefs' by taking the new examples of input and output into account, where it infers the shared latent concepts across these examples. The paper shows the ability of models like Transformers and LSTMs to perform ICL and its ability to infer shared latent concepts. In Question Answering problems, we often deal with In-Context Learning when the context needed to answer the problem, is provided to the LLM.

In order to evaluate the performance of LLMs, we need questions that test the model's ability to read multiple paragraphs in order to get the final answer. (Ho et al., 2020) worked on creating a dataset that contains a wide variety of questions to help test the models on this aspect. The authors of the paper exploited the structured format that is inherently present in Wikipedia — the source for their data — and used logical rules to create questions that are challenging and require multi-hop reasoning.

2.2 Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) has experienced rapid development due to its ability to enhance the performance of LLMs and reduce hallucinations (Gao et al., 2024; Lewis et al., 2021). RAGs integrate external knowledge into the text generation. They use semantic similarity algorithms to retrieve relevant documents from external databases, enabling them to continuously update the knowledge of LLMs and incorporate domain-specific information (Gao et al., 2024). It has been proven that these characteristics make RAGs suitable for handling knowledge-intensive NLP tasks, such as open-domain question answering (Gao et al., 2024; Lewis et al., 2021).

2.3 Dense retrieval

Traditional sparse retrieval methods, like BM25, are widely used (Robertson and Zaragoza, 2009), but they suffer from vocabulary mismatches. In a case, where two words are semantically related, they will not be recognized as the same when using sparse retrieval. Dense Retrieval (DR) addresses this problem by introducing learned representations in a vector space. (Karpukhin et al., 2020) introduced Dense Passage Retrieval (DPR) which outperforms traditional sparse methods in Open-Domain QA. DPR allows for semantic-based retrieval instead of exact keyword matches.

3 Method

For all experiments, the library Dexter-CQA (V. et al., 2024) is used in combination with the "Flan T5-xl" model (Chung et al., 2022).

3.1 Setup

The dataset that was used in all experiments is called 2WikiMultiHopQA (Ho et al., 2020). It consists of questions, contexts and answers. To be able to answer these questions, a system first needs to retrieve multiple pieces of information, and after do multiple steps of reasoning to get to the right answer (Mavi et al., 2024). Later on, when we talk about evidence, we refer to the relevant contexts needed to answer the question.

For the LLM, the "FLAN-T5-xl" model was chosen (Chung et al., 2022) because of the low compute requirements. In order to evaluate the results for each experiment, the Exact Match (EM) implementation from Dexter-CQA (V. et al., 2024) is used. This implementation executes normalization of the text before comparing if they are exactly the same. These normalizations include transforming to lower case, removing punctuation, removing articles and removing redundant white space. The prompt that was used on our GitHub repository (Karskens et al., 2024).

3.2 Contriever

Contriever is a model that was introduced by Facebook in their paper (Izacard et al., 2021) "Unsupervised Dense Information Retrieval with Contrastive

Learning". Instead of encoding the document and query together in document-query pairs, a Contriever directly encodes the document and query separately using a bi-encoder architecture and then computes the dot product of the two. This makes it more scalable as the documents and the query need not be re-encoded for each query which was done in the case of traditional dense retrievers.

We use the Contriever directly implemented in the Dexter-CQA project and use cosine similarity as the metric to get top-k documents for each query. From this we get a response list that we use to query the relevant evidences (documents) for each question.

For our Contriever setup, we use few-shot learning while querying so that the FLAN-T5 model can output the final answer in a standard format. We run the experiments using the top-5, top-3 and top-1 relevant document provided as evidences to the LLM and calculate the exact match metric between the generated answer and the gold standard answer present in dev.json file of the 2WikiMultiHopQA dataset (Ho et al., 2020).

3.3 Oracle contexts

For this experiment, we directly use the oracle contexts provided in the dev.json file as part of the 2WikiMultiHopQA dataset (Ho et al., 2020). The dataset contains documents for each query that will be used in our RAG setup. We repeat the same experiment as last time, using the FLAN-T5 LLM and using the top-5, top-3 and top-1 relevant document provided as the evidence to the model, this time getting the relevant documents from the oracle contexts.

3.4 Random negative documents

Documents classified as random have no relationship to the query. Therefore, adding random documents to the retrieval process can be seen as injecting noise. It is important to note that these random documents do not contain the answer to the query either. Research shows that when injecting noise in a right manner, it can improve the performance of the model (Yu et al., 2022). Introducing noise into the query can cause the model to focus more on the important signals amidst the noise.

To implement this, we made use of the already existing corpus containing all the documents. When random sampling, it is important that the data we sample has the following two properties:

1. The document cannot contain the answer
2. The document cannot already be retrieved

We run three experiments in order to evaluate whether random sampling is beneficial. To compare effectively with the results obtained without random sampling, we run with it with the following k-values: 1,3,5. After retrieving the top-k documents, we inject noise into the evidence by selecting a ratio and sampling according to that ratio from the corpus. After sampling of the documents, we append it to the top-k documents retrieved and utilise this newly constructed evidence as generator input.

3.5 Hard negative documents

Hard negative context documents are those close to the query in terms of semantic similarity, when measured in the vector space, but are not actually useful to answer the query. In this research, our aim is to investigate the impact and importance of using hard negative documents in RAG systems for complex question answering.

To achieve that, we use the Contriever retrieval model on the corpus to retrieve a set of top 100 document candidates based on the Cosine Similarity score. From the retrieved candidate set, we then identify which documents are the known ground truth documents, actually containing the correct evidence to answer the question, and remove them from the set. From the remaining set containing only non-gold context documents, we can now pick the most relevant but not ground-truth documents as hard negatives.

To evaluate the injection of negative context documents into the RAG system, we use the top k=1, 3 and 5 relevant evidences extracted in [3.2]. In each k, we inject the hard negative documents in different ratios, ensuring that the hard negatives are not already in the list of the top-k evidences. Finally, we use the different injected inputs to prompt FLAN-T5-xl and evaluate the results.

3.6 ADORE

Instead of using Contriever [3.2] to extract the contexts, we also conduct the same experiment with a model called ADORE (Zhan et al., 2021). ADORE is a model that is specifically optimized with hard negatives and is trained by using dynamic hard negatives instead of static hard negatives. This means that the hard negatives are dynamically changed

during training instead of the static hard negatives that are retrieved once before or during the training process. Hard negatives are defined as the top-k relevant documents. To retrieve the dynamic hard negatives during training the researchers do the following. A new batch of queries is encoded using the model after every training round. These query embeddings are used to retrieve the currently relevant documents from which the top documents serve as the dynamic hard negatives. Since ADORE only trains the query encoder, another model is needed to encode the documents. In the experiments, we used the STAR algorithm which was also introduced in the same paper. In the paper, they show theoretically that using normally static hard negative sampling is not stable. The STAR algorithm is presented to combat this by using random negatives that should stabilize the training process.

An experiment was conducted to test whether using ADORE results in better downstream answer generation performance. A retriever is trained with ADORE. The first 1200 queries of the training file were selected from the 2WikiMultiHopQA dataset (Ho et al., 2020) to be used as training data. The IDs can be found in our GitHub repository (Karskens et al., 2024). The STAR model, provided by the researchers of ADORE (Zhan et al., 2021), is used to calculate the document embeddings and is used as a start model for ADORE. The maximum number of tokens for documents was set to 512 tokens. So, if documents surpass this, they are truncated to a maximum of 512 tokens. ADORE will further improve the STAR model by training its query-encoder. The training was done for 6 epochs and used the default settings from the training script provided by the researchers (Zhan et al., 2021).

The ADORE retriever is used to get the top-k relevant documents. The documents that are retrieved are used as context in the prompt for the Flant5-xl model. The outputs are compared with the real answers by ExactMatch from Dexter-CQA (V. et al., 2024). The performance of this method is compared to the performance of the baseline Contriever experiment in section 4.5.

3.7 Testing Set

Since running the Flant5-xl on all the dev dataset questions is computationally expensive for the hardware available to us, the testing set consists of the first 100 questions taken from the dev dataset of 2WikiMultiHopQA (Ho et al., 2020).

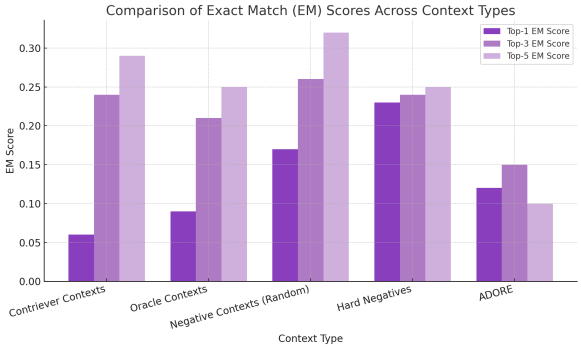

Figure 1: Comparison of EM scores

4 Results

The goal of the experiments is to analyse the impact of different type of contexts, such as gold, related and negative on RAG systems in the context of Open Domain Complex QA. The performance of the RAG system is evaluated using Exact Match (EM) scores. The results reported in Table 1 for random negative contexts and hard negatives are the best performing ratios. All experiments are conducted on the same dataset described in the Method section.

4.1 Experimental setup

All experiments were run on a laptop with 32GB RAM and RTX3070ti. In

4.2 RQ1: How do negative contexts impact downstream answer generation performance?

4.2.1 Effect of Mixing Contexts: Noise

In table 2, we can see the different experiments run to evaluate the effect of injecting random noise into the evidence. The experiments are divided into top-1, top-3 and top-5 results retrieval. Different ratios were utilised so that a comparison could be made between them. From the results, we notice the following trend: adding random documents improves the EM score of the model. This trend can be seen for the top-1, top-3 and top-5 retrieval approaches insinuating that adding random documents does indeed improve the performance over only using Contriever contexts or oracle contexts.

The different ratios used in the experiments are informative as it shows that more random documents does not necessarily improve the performance. Specifically for top-3 retrieval, we can see

Context Type	Top-1 EM Score	Top-3 EM Score	Top-5 EM Score
Contriever Contexts	*0.06*	*0.24*	*0.29*
Oracle Contexts	*0.09*	*0.21*	*0.25*
Negative Contexts (Random)	*0.17*	***0.26***	***0.32***
Hard Negatives	***0.23***	*0.24*	*0.25*
ADORE	*0.12*	*0.15*	*0.10*

Table 1: Exact Match (EM) scores for different context types across Top-1, Top-3, and Top-5 retrievals. The results for random negative contexts and hard negatives are using the best performing ratios.

that injecting two random documents improves the EM score, but injecting 3 random documents decreases the performance of the model. Showcasing that the injection of too many random documents might confuse the model and decrease the performance.

4.2.2 Effect of Mixing Contexts: Hard Negatives

Table 2 presents the results of injecting hard negative contexts into the evidence. The results showcase that based on the mixing strategy (number of relevant documents compared to the number of hard negatives), the impact on the performance of the LLM varies. As the number of relevant documents increases, the benefit of injecting hard negatives diminishes. Specifically, injecting noise to the top-1 relevant document, improves the performance of the LLM, compared to only providing the relevant document. However, incorporating hard negatives alongside the top-3 or top-5 relevant documents either maintains the EM score at its current level or causes it to decline, depending on the quantity of hard negatives injected.

When we contrast the results of mixed prompting with prompting using the oracle contexts only, we observe that the LLM can sometimes perform better with a mix of relevant documents and hard negatives. In contrast, when the number of oracle documents increases (k=5), the model matches or outperforms the use of the injected input, depending on the injection ratio.

4.3 RQ2: Are negative contexts more important for answer generation than related contexts?

As can be seen in table 1, the addition of random noise improves the model performance implying that random negative contexts are indeed important for satisfying model performance. This indicates that adding random noise improves the ability of the model to generalise better. Furthermore, it allows the model to differentiate between relevant and non-relevant, enhancing its robustness.

In contrast, in most cases, injecting hard negatives does not improve model's performance compared to related contexts (see table 2). This indicates that while hard negatives introduce the model to challenging distinctions, it also leads to confusion which hinders the ability of the LLM to perform well.

4.4 RQ3: Does providing only gold contexts deteriorate the performance compared to mixing with other negative or related contexts?

Yes indeed, providing only gold contexts deteriorates the performance compared to mixing with other negative or related contexts. Taking a look at the figures from table 1 we can see that providing only oracle contexts does not lead to the best results. There can be several reasons for this — the primary reason being that while oracle contexts do contain ground truth, for several questions this involves contexts that may not be important for the final answer. Therefore, while taking top-k documents present in oracle contexts, we might have a lot of unnecessary information that is not required for the final answer and we might be missing relevant documents. Whereas, when we use Contriever we might be able to fetch the relevant documents which improves the final score. Even noise and hard negatives help the LLM more as is illustrated in the previous sections.

Of course, the results might be very different if we use all the contexts present in the oracle and/or only use the relevant evidences. This can be explored in a future work.

4.5 Contriever vs. ADORE

In Table 1, it can be seen that the ADORE model (Zhan et al., 2021) performs better by 6 percent compared to the Contriever model [3.2] when looking at the top 1 documents. However, in Table 1

Context Mix	Random Negatives EM Score	Hard Negatives EM Score
1 Relevant, No Noise	*0.06*	*0.06*
1 Relevant, 1 Noise	*0.17*	*0.23*
3 Relevant, No Nose	*0.24*	*0.24*
3 Relevant, 1 Noise	*0.25*	*0.23*
3 Relevant, 2 Noise	*0.26*	*0.24*
3 Relevant, 3 Noise	*0.24*	*0.22*
5 Relevant, No Noise	*0.29*	***0.29***
5 Relevant, 1 Noise	***0.32***	*0.24*
5 Relevant, 3 Noise	*0.24*	*0.25*
5 Relevant, 5 Noise	*0.18*	*0.21*

Table 2: Exact Match (EM) scores for different Noise-Mixing strategies

one can see that for the top 3 and top 5, the Contriever obtains a higher score. For the Contriever model, a steep increase in the EM score can be seen when going from the top 1 in Table 1 to the top 3 documents. From the top-3 to the top-5 the Contriever EM score only increases by 5 percent. For the ADORE model, the EM score increases a bit from the top 1 to the top 3 but actually drops by 5 percent from the top 3 to the top 5. The ADORE EM scores are also the lowest compared to all the other retrieval methods that were tested.

5 Qualitative Analysis

5.1 Question 1

Question: Do both films, Dial M For Murder and Marius (1931 Film), have the directors who are from the same country?

Answer: Yes

All experiments returned no. This is a reasonably complex question as it requires the LLM to extract the directors of both films and then the country they came from and compare the two. This involves 4 relevant ground truth documents which were not retrieved in any of the experiments we ran except oracle contexts but the LLM was unable to answer correctly nevertheless.

5.2 Question 2

Question: Which film has the director who died earlier, Captain Kidd And The Slave Girl or Relax Freddie?

Answer: Captain Kidd And The Slave Girl

Contriever was not able to get the relevant contexts, therefore the LLM resorted to guessing the answer which luckily turned out to be correct. ADORE was also not able to retrieve the correct contexts, however, turned out to only give the correct answer in the top 3 and top 5 experiments.

Random negatives did improve the performance on this question, as both top-1 and top-5 retrieval produced the right result. Indicating that generalisation capabilities are improved when injecting random noise into the evidences. However, injecting hard negatives into the relevant evidences confused the model, giving the wrong answer in all experiments. This showcases that the LLM struggles to identify the correct answer when documents with high similarity but not actually relevant are included in the context.

6 Discussion

6.1 Why are random negative contexts better than oracle contexts?

From the results we can see an improvement in model performance when comparing oracle contexts to random negative contexts. This can be explained by the following: by including random noise, the model is able to generalise better and therefore, able to answer unseen questions better. Research on robustness of RAG systems confirms our hypothesis. Oracle contexts provide the most relevant passages for answering the query, but this also allows for the model to be lazy (Yu et al., 2022). Over-reliance on oracle contexts can lead to not proper multi-hop reasoning being executed. Therefore, introducing random negative contexts into the evidences allows for the model to generalise and perform true multi-hop reasoning.

6.2 Why Exact Match (EM) is not a representative metric?

EM requires the output of the LLM to exactly match the ground truth given. Even if the generated

answer is semantically correct, if the phrasing is different or if the words are ordered differently, EM will mark the answer as wrong. Especially, considering that the model requires multi-step reasoning, when piecing back the retrieved information there is a high variability in phrasing. Furthermore, it does not take into account partially right answers. Additionally, the EM score often does not match human judgment. An example from our experiment is the following:

- **Ground Truth**: "United States"
- **Model Output**: "American"

Even though both answers are equivalent, the model will output an EM score of 0.

6.3 Why were the documents provided by Contriever better as compared to the oracle contexts?

From the results we can see that Contriever [3.2] top-k documents perform better for top-3 and top-5 relevant documents as compared to the oracle contexts. This can be explained by the following: oracle contexts contain a lot of documents that may/may not be relevant for the final answer. Therefore, when we fetch the top-k documents from oracle contexts we might miss out on relevant documents and give a lot of unnecessary information to the LLM which might confuse it. Whereas, the Contriever will fetch more relevant documents as it uses cosine similarity to get top-k relevant documents. This makes the experiments run with Contriever documents perform better as compared to oracle contexts.

6.4 Why does the performance decrease for hard negatives when compared to random negatives?

Since hard negatives are close in the semantic space to the gold contexts, the LLM might have greater difficulty, when closely related contexts are provided, in identifying the right information to answer the question. Instead, when prompting with random negatives, the LLM understands the gold contexts better — the one required for generating the final answer — and is able to distinguish it from the random negative ones which are not as close in the semantic space.

6.5 Low Exact Match (EM) scores

The relatively low EM scores observed in our experiments can be linked to the choice of LLM. The Flant-T5-xl is efficient for computation but is a relatively small model when compared to bigger LLMs, like ChatGPT. Given its size, the Flant model has limitations in its capabilities. As can be seen in (V. et al., 2024), there is a significant difference in performance between the results reported in the original paper of Dexter-CQA and ours due to using different LLMs.

6.6 Why does ADORE perform worse when compared to the other experiments?

The ADORE (Zhan et al., 2021) retrieval model performs way worse compared to the other retrieval methods. This could have multiple reasons. One of the reasons could be that the model was not trained with a large enough dataset. The dataset only contained 1200 questions which could be too small. Another reason could be the number of epochs that are set to 6. Performance could have improved when training for more than 6 epochs.

6.7 Gold context positioning

Another interesting thing could be the positioning of the context of the question. In our experiments, the the top context was always furthest away from the question and the last context in the top-k contexts was always closest. In the field, there are findings that the positioning of the correct or gold context is important for the performance (Cuconasu et al., 2024). The researchers found that when context is placed in the middle it would result in the lowest accuracy. Higher accuracy, when placing the context far away and the highest accuracy is received when placing the context closest to the query. This phenomenon could potentially partly explain why ADORE (Zhan et al., 2021) has a higher score for the top 1 but lower for the top 3 and top 5. It could be that ADORE is better at retrieving the correct document in the first position or higher in the ranking than Contriever (Izacard et al., 2021). Since, for our experiments, the lowest ranked document is always closest to the question it could be that for the top 3 and top 5, the right context was in position 1 or 2 for ADORE, and for Contriever it could be in position 3 and 5. This would improve the performance of Contriever even though the ADORE model could have done a better job ranking the relevance of the documents.

7 Conclusion

To summarize, it is possible for Contriever retrieved documents to outperform oracle contexts because a Contriever can find more related documents as compared to blindly using top-k fetched oracle contexts. Of course, maybe this pattern will not remain true if we use all the ground truth documents present in the dataset which can be explored in a further study. Through these experiments we can safely say that closely related documents that are not relevant for the final answer might act as a source of confusion for the LLM and it might predict the wrong answer — this part is explored more in the hard negatives experiment.

Furthermore, it is evident from the results that the addition of random noise in certain ratios does improve the performance of the model for all top-k. Including random noise improves the model's ability to generalise and leads to better results. However, the results showcase that incorporating hard negative context documents in relevant documents ends up decreasing the performance of the RAG in most cases. The reason for that may be that the LLM finds difficulty in distinguishing which context is more relevant to answer the question correctly because of the high similarity of the hard negatives with the query and the correct evidences.

Additionally, the results of the ADORE experiment showed that our retrieval model that was trained with dynamic hard negatives is not better than using Contriever or random negative contexts. Potential reasons for this were the size of the training set and the influence of the positioning of the gold context in the prompt.

Further research can be conducted with different metrics and/or different LLMs to examine whether the performance can be improved.

References

Hyung Won Chung, Le Hou, Shayne Longpre, Barret Zoph, Yi Tay, William Fedus, Eric Li, Xuezhi Wang, Mostafa Dehghani, Siddhartha Brahma, Albert Webson, Shixiang Shane Gu, Zhuyun Dai, Mirac Suzgun, Xinyun Chen, Aakanksha Chowdhery, Sharan Narang, Gaurav Mishra, Adams Yu, Vincent Zhao, Yanping Huang, Andrew Dai, Hongkun Yu, Slav Petrov, Ed H. Chi, Jeff Dean, Jacob Devlin, Adam Roberts, Denny Zhou, Quoc V. Le, and Jason Wei. 2022. Scaling instruction-finetuned language models.

Florin Cuconasu, Giovanni Trappolini, Federico Siciliano, Simone Filice, Cesare Campagnano, Yoelle Maarek, Nicola Tonellotto, and Fabrizio Silvestri. 2024. The power of noise: Redefining retrieval for rag systems. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*, SIGIR 2024, page 719–729. ACM.

Yunfan Gao, Yun Xiong, Xinyu Gao, Kangxiang Jia, Jinliu Pan, Yuxi Bi, Yi Dai, Jiawei Sun, Meng Wang, and Haofen Wang. 2024. Retrieval-augmented generation for large language models: A survey.

Xanh Ho, Anh-Khoa Duong Nguyen, Saku Sugawara, and Akiko Aizawa. 2020. Constructing a multi-hop QA dataset for comprehensive evaluation of reasoning steps. In *Proceedings of the 28th International Conference on Computational Linguistics*, pages 6609–6625, Barcelona, Spain (Online). International Committee on Computational Linguistics.

Gautier Izacard, Mathilde Caron, Lucas Hosseini, Sebastian Riedel, Piotr Bojanowski, Armand Joulin, and Edouard Grave. 2021. Unsupervised dense information retrieval with contrastive learning. *arXiv preprint arXiv:2112.09118*.

Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Yejin Bang, Andrea Madotto, and Pascale Fung. 2022. Survey of hallucination in natural language generation. *CoRR*, abs/2202.03629.

Vladimir Karpukhin, Barlas Oguz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. 2020. Dense passage retrieval for open-domain question answering. *arXiv preprint arXiv:2004.04906*.

Samuel Karskens, Elvira Antonogiannaki, Sankalp Sagar, and Vithursika Vinasiththamby. 2024. Nlp project. Accessed: January 22, 2025.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2021. Retrieval-augmented generation for knowledge-intensive nlp tasks.

Vaibhav Mavi, Anubhav Jangra, and Adam Jatowt. 2024. Multi-hop question answering.

Stephen Robertson and Hugo Zaragoza. 2009. The probabilistic relevance framework: Bm25 and beyond. *Foundations and Trends in Information Retrieval*, 3(4):333–389.

Venktesh V., Deepali Prabhu, and Avishek Anand. 2024. Dexter: A benchmark for open-domain complex question answering using llms.

Sang Michael Xie, Aditi Raghunathan, Percy Liang, and Tengyu Ma. 2021. An explanation of in-context learning as implicit bayesian inference. *arXiv preprint arXiv:2111.02080*.

Youngjae Yu, Jiwan Chung, Heeseung Yun, Jack Hessel, Jae Sung Park, Ximing Lu, Prithviraj Ammanabrolu, Rowan Zellers, Ronan Le Bras, Gunhee Kim, and Yejin Choi. 2022. Multimodal knowledge alignment with reinforcement learning. *arXiv preprint arXiv:2205.12630*.

Jingtao Zhan, Jiaxin Mao, Yiqun Liu, Jiafeng Guo, Min Zhang, and Shaoping Ma. 2021. Optimizing Dense Retrieval Model Training with Hard Negatives. In *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 1503–1512, Virtual Event Canada. ACM.